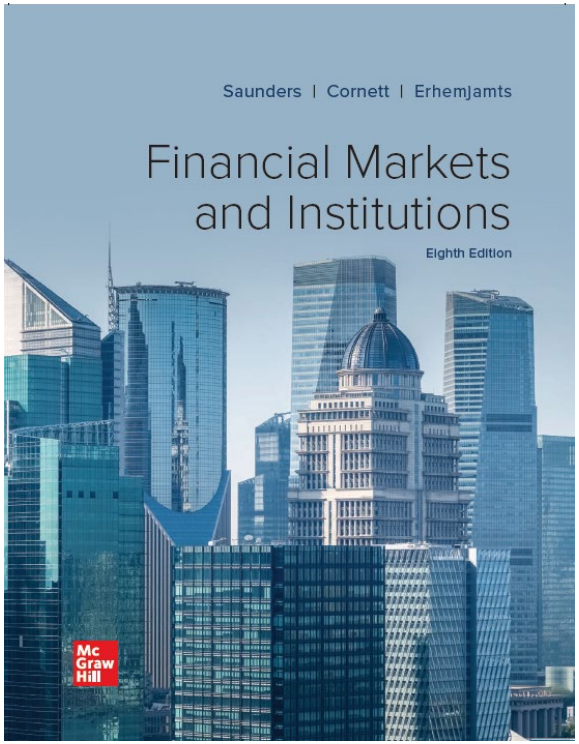
Chapter Eight

Stock Markets



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The Stock Markets: Chapter Overview

- Corporate stock serves as a source of financing for firms, in addition to debt financing or retained earnings financing
- Secondary stock markets are the most closely watched and reported of all financial security markets
- Stockholders are the legal owners of a corporation
  - Have a right to share in the firm's profits (through dividends) after the payment of interest to bond holders and taxes
  - Have a residual claim on the firm's assets
  - Have limited liability
  - Have voting rights (e.g., to elect board of directors)

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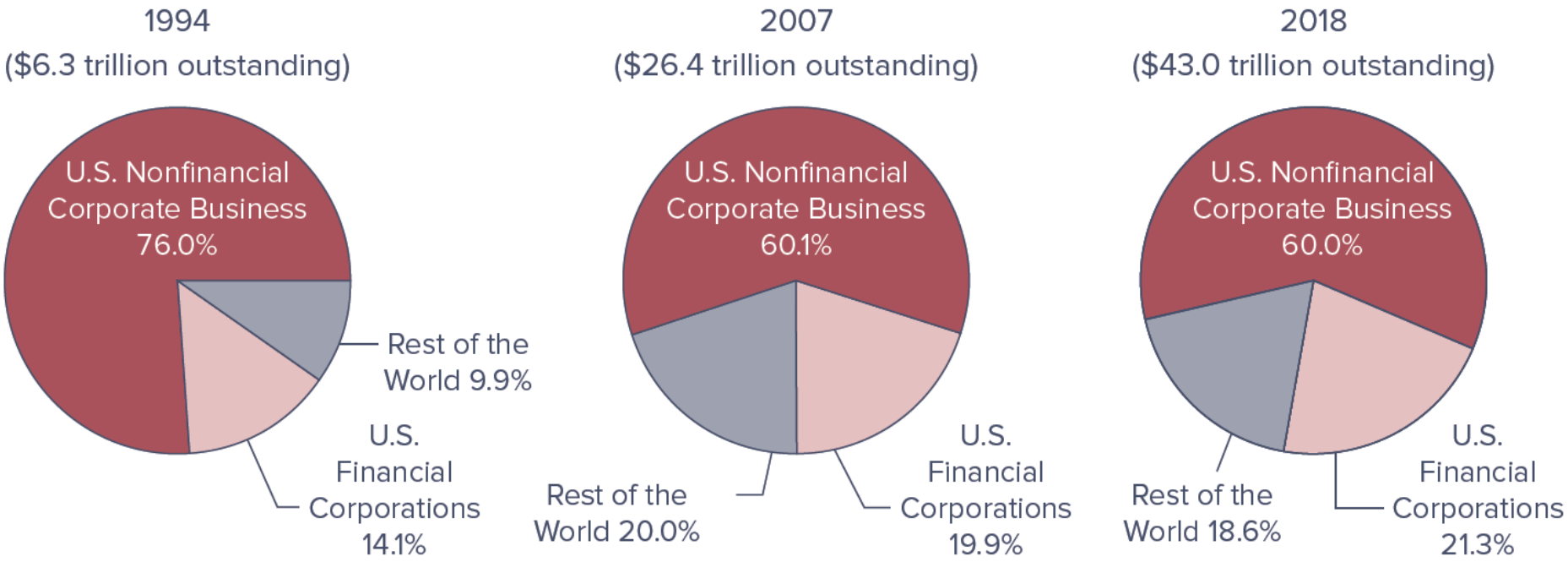
Market Value of Common Stock Outstanding, by Type of Issuer



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Stock Market Securities: Common Stock

- Two types of corporate stock exist:
  1. Common stock
  2. Preferred stock
- All public corporations issue common stock, but few issue preferred stock
- **Common stock** is the fundamental ownership claim in a public or private corporation, and many characteristics differentiate it from other types of securities:
  - Discretionary dividend payments
  - Residual claim status
  - Limited liability
  - Voting rights

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Common Stock: Dividends

- **Dividends** are discretionary, and are thus not guaranteed
  - Payment and size of dividends are determined by the board of directors of the issuing firm
- Dividends are taxed twice – once at the firm level and once at the personal level
  - May partially avoid this double taxation effect by holding stocks in growth firms that reinvest most of their earnings to finance growth rather than paying larger dividends

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Common Stock: Dividends (Continued)

- The return to a stockholder over a period *t*-1:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} + \frac{D_t}{P_{t-1}}$$

where

$$P_t = \text{Stock price at time } t$$

$$D_t = \text{Dividends paid over time } t-1 \text{ to } t$$

$$\frac{P_t - P_{t-1}}{P_{t-1}} = \text{Capital gain over time } t-1 \text{ to } t$$

$$\frac{D_t}{P_{t-1}} = \text{Return from dividends over time } t-1 \text{ to } t$$

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Common Stock: Residual Claim

- Common stockholders have the lowest priority claim in the event of bankruptcy (i.e., they have a **residual claim)**
- Only after all senior claims are paid are common stockholders entitled to what assets of the firm are left
  - Senior claims may be payments owed to creditors such as the firm's employees, bond holders, the government (taxes), and preferred stockholders
- Residual claim feature associated with common stock makes it riskier than bonds as an investible asset

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Common Stock: Limited Liability

- **Limited liability** implies that common stockholder losses are limited to the amount of their original investment in the firm if the company's asset value falls to less than the value of the debt it owes
  - In contrast, sole proprietorship or partnership stock interests mean the stockholders may be liable for the firm's debts out of their total private wealth holdings if the company experiences financial difficulties

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Common Stock: Voting Rights

- Common stockholders control the firm's activities indirectly by exercising their *voting rights* in the election of the board of directors
- Typical voting rights arrangement is to assign one vote per share of common stock
- Some firms are organized as **dual-class firms**, where two classes of common stock are outstanding, with different voting and/or dividend rights for each class
- Two methods of electing a board are generally used:
  1. **Cumulative voting**
  2. Straight voting

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Common Stock: Proxy Votes

- Most shareholders do not attend annual meetings
- A **proxy** is a voting ballot sent by a corporation to its stockholders
  - When returned to the issuing firm, a proxy allows stockholders to vote by absentee ballot or authorizes representatives of the stockholders to vote on their behalf
- By the 2010s, virtually all U.S. firms were putting proxy statements online and allowing votes to be cast via the Internet

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Preferred Stock

- **Preferred stock** is a hybrid security that has characteristics of both bonds and common stock
  - Similar to common stock in that it represents an ownership interest in the issuing firm, but like a bond it pays a fixed periodic (dividend) payment
- Dividends are generally fixed (paid quarterly)
- Preferred stockholders generally do not have voting rights in the firm, but most stock may be converted to common stock at any time the investor chooses

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Preferred Stock (Continued)

- Typically, preferred stock is nonparticipating and cumulative
  - **Nonparticipating preferred stock** means the dividend is fixed regardless of any increase of decrease in the issuing firm's profits, while **participating preferred stock** means actual dividends paid in any year may be greater than promised dividends
  - **Cumulative preferred stock** means that any missed dividend payments go into arrears and must be made up before any common stock dividends can be paid, while dividends of **noncumulative preferred stocks** do not go into arrears and are never paid

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Primary Stock Markets

- **Primary stock markets** are markets in which corporations raise funds through *new* issues of stocks
  - Most primary market transactions go through investment banks
- Investment bank can conduct a primary sale using either a firm commitment or best efforts underwriting basis
  - In firm commitment underwriting, the investment bank guarantees the corporation a price for the newly issued securities
  - Best efforts underwriting occurs when the underwriter does not guarantee a price to the issuer
- A **syndicate** is a group of investment banks working in concert to sell and distribute a new issue; the lead banks in the syndicate is the **originating house**

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Primary Market Stock Transaction

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Primary Stock Markets (Continued)

- An **initial public offering (IPO)** is the *first* public issue of a financial instrument by a firm
- A **seasoned offering** is the sale of *additional* securities by a firm whose securities are currently publicly traded
  - **Preemptive rights** give existing stockholders the ability to maintain their proportional ownership
- Registration of a stock can be a lengthy process
  - A **red herring prospectus** is a preliminary version of the prospectus that describes a new security issue
  - **Shelf registration** allows firms that plan to offer multiple issues of stock over a two-year period to submit one registration statement (i.e., master registration statement)

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Secondary Stock Markets: Stock Exchanges

- **Secondary stock markets** are the markets in which stocks, once issued, are traded by investors
  - In secondary market transactions, funds are exchanged, usually with the help of a securities broker or firm acting as an intermediary between the buyer and seller of the stock
  - Original issuer of the stock is not involved in this transfer of stocks or funds
- Two major U.S. stock markets are the following:
  1. New York Stock Exchange Euronext (NYSE Euronext)
  2. National Association of Securities Dealers Automated Quotation (NASDAQ) system

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Secondary Stock Markets: The New York Stock Exchange

- Worldwide, the NYSE Euronext is the most well known of all the organized exchanges
  - Exchange was created by the merger of the NYSE Group, Inc. and Euronext N.V. on April 4, 2007
  - World leader for listings, trading in cash equities, equity and interest rate derivatives, bonds, and the distribution of market data
  - First to create a truly global stock market

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Trading Process

- All transactions occurring on the NYSE occur at a specific place on the floor of the exchange (**trading post**), and each stock is assigned a special market maker (**a specialist**)
  - Specialists often organize themselves as firms due to large amount of capital needed to serve the market-making function
- Three types of transactions can occur at a given post:
  1. Brokers trade on behalf of customers at the “market” price (market order)
  2. Limit orders are left with a specialist to be executed
  3. Specialists transaction for their own account

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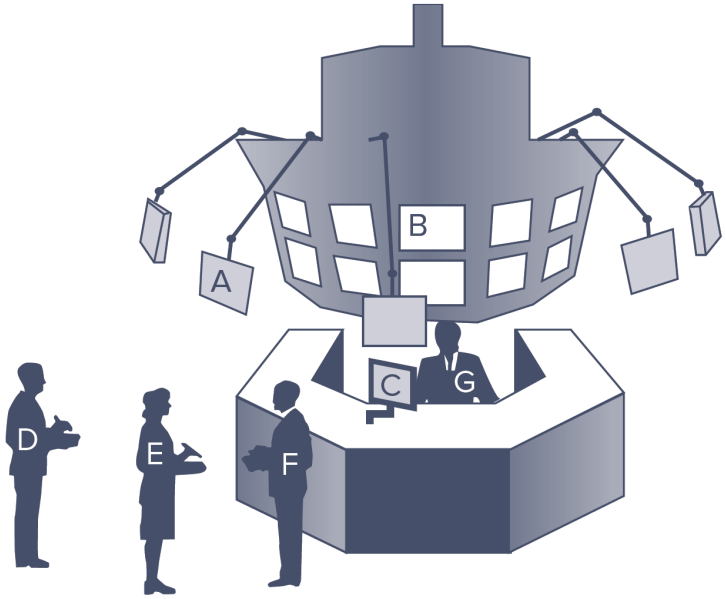
Trading Process (Continued)

- Majority of orders sent to brokers are of two types:
  - A **market order** is an order to transact at the best price available when the order reaches the post
  - A **limit order** is an order to transact at a specified price



A Typical Trading Post

A: Post information display
B: Flat-panel display screen
C: Point-of-sale workstation
D: Broker
E: Broker
F: Designated market maker (DMM)
G: DMM trading assistant

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U.S. Stock Gainers, February 21, 2020

(1) Name	(2) Volume	(3) List	(4) Chg	(5) %Chg
Trans World Entertainment Corp. (TWMC)	5.2M	5.28	1.78	50.86
Celldex Therapeutics Inc. (CLDX)	8.6M	3.07	0.80	35.24
Ducommun Inc. (DCO)	939.7K	56.95	14.16	33.09
Eldorado Gold Corp. (EGO)	20.2M	9.93	2.41	32.05
Avadel Pharmaceuticals PLC ADR (AVDL)	5.8M	9.40	2.28	32.02
Genprex Inc. (GNPX)	11.7M	5.33	0.95	21.69
American Software Inc. Cl A (AMSWA)	632.3K	19.20	3.30	20.75
IGM Biosciences Inc. (IGMS)	351.4K	60.69	10.22	20.25
Dropbox Inc. (DBX)	48.3M	22.45	3.73	19.96
China Xiangtai Food Co. Ltd. (PLIN)	293.3K	3.07	0.49	18.99
Oncolytics Biotech Inc. (ONCY)	1.9M	2.63	0.40	17.94
National CineMedia Inc. (NCMI)	2.7M	9.37	1.37	17.13
California Resources Corp. (CRC)	9.1M	7.59	1.09	16.77
Microbot Medical Inc. (MBOT)	1.4M	10.81	1.54	16.61
Fang Holdings Ltd. ADR (SFUN)	219.7K	2.57	0.36	16.29
Universal Electronics Inc. (UEIC)	328.0K	51.80	6.82	15.16
Baudax Bio Inc. (BXRX)	7.9M	9.16	1.16	14.51

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The NASDAQ and OTC Market

- Securities not sold on one of the organized exchanges are traded over the counter (OTC)
  - OTC markets do not have a physical trading floor; rather, transactions are completed via an electronic market
- NASDAQ was the world's first electronic stock market
  - Primarily a dealer market, where dealers are the market makers who stand ready to buy or sell particular securities
  - In contrast to the NYSE, NASDAQ is a negotiated market (e.g., quotes from several dealers are usually obtained before a transaction is made)
  - Small Order Execution System (SOES) provides automatic order execution for individual traders with orders of less than or equal to 1,000 shares

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Choice of Market Listing

- Firms listed with the NYSE Euronext must meet the listing requirements of the exchange
  - Requirements are extensive
- Reasons a NYSE listing is attractive to a firm:
  - Improved marketability of the firm's stock
  - Publicity for the firm
  - Improved access to the financial markets
- Firms that do not meet the requirements of the NYSE Euronext exchange listings trade on the NASDAQ
  - Most NASDAQ firms are smaller, of regional interest, or unable to meet the listing requirements of the organized exchanges
  - Over time, many NASDAQ firms apply for NYSE listing

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Electronic Communications Networks and Online Trading

- Major stock markets currently open at 9:30 am eastern time and close at 4:00 pm eastern time
  - Extended-hours trading involves any securities transaction that occurs outside these regular trading hours
  - Most extended-hours trading is processed through computerized alternative trading systems (ATSs), also known as electronic communication networks (ECNs) such as NYSE Arca or Instinet
- ECNs are computerized systems that automatically match orders between buyers and sellers and serve as an alternative to traditional market making and floor trading

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Stock Market Indexes

- A stock market index is the composite value of a group of secondary market-traded stocks
  - Dow Jones Industrial Average (DJIA) is the most widely reported stock market index and includes the values of 30 large corporations selected by the editors of The Wall Street Journal
    - Dow indexes are *price-weighted averages*
  - NYSE Composite Index includes all NYSE-listed common stocks
    - NYSE is a *value-weighted index*
  - S&P 500 Index consists of the stocks of the top 500 of the largest U.S. corporations listed on the NYSE and the NASDAQ
  - NASDAQ Composite Index consist of stocks traded through NASDAQ that are industrials, banks, and insurance companies

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Stock Market Indexes (Continued)

- Wilshire 5000 Index is the broadest stock market index and possibly the most accurate reflection of the overall stock markets
  - Contains virtually every stock that meets three criteria:
    1. Firm is headquartered in the U.S.
    2. Stock is actively traded in a U.S.-based stock market
    3. Stock has widely available price information (which rules out the smaller OTC stocks from inclusion)
  - Though the index started with 5,000 firms, because of firm delistings, privatizations, and acquisitions it currently includes just 3,618 stocks
  - Value-weighted index

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Other Issues Pertaining to Stock Markets: Economic Indicators

- Stock market indexes might be used to forecast future economic activity
  - An increase (decrease) in stock market indexes today potentially signals the market's expectation of higher (lower) corporate dividends and profits and, in turn, higher (lower) economic growth
- Stock prices are one of the 10 variables included in the index of leading economic indicators used by the Federal Reserve as it formulates economic policy

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Other Issues Pertaining to Stock Markets: Market Efficiency

- **Market efficiency** refers to the speed with which financial security prices reflect unexpected news
  - *Weak form market efficiency* concludes that investors cannot make more than the fair (required) return using information based on historic price movements
    - Empirical research suggests markets are weak form efficient
  - According to *semistrong form market efficiency*, investors cannot make more than the fair (required) return by trading on public news releases
    - Financial markets have generally been found to immediately reflect information from news announcements
  - *Strong form market efficiency* states that stock prices fully reflect all information about the firm, both public and private
    - Implies that there is no set of information that allows investors to make more than the fair (required) rate of return on a stock

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International Aspects of Stock Markets

- U.S. stock markets are the world's largest
  - European markets grew in importance during the 2000s, as a result of implementing the euro, a common currency, in 2002
- International stock markets are attractive to investors because some risk can be eliminated by holding stocks issued by corporations in foreign countries
- International diversification can also introduce risk in the following manners:
  - Information about foreign stocks is less complete and timely than that for U.S. stocks
  - Foreign exchange risk
  - Political (sovereign risk)

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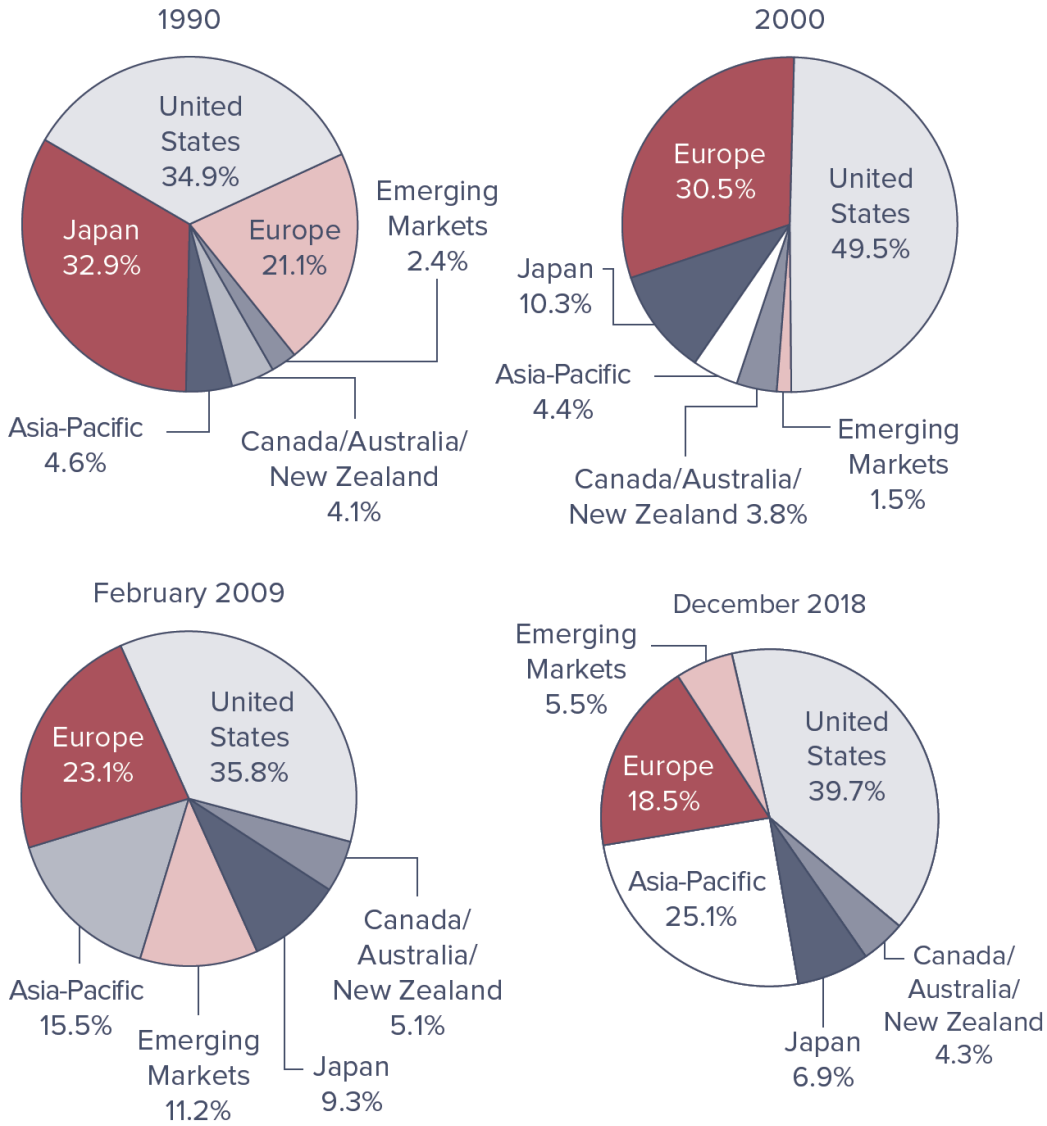
Worldwide Stock Market Capitalization



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American Depository Receipts (ADRs)

- An ADR is a certificate that represents ownership of a foreign stock
  - Typically created by a U.S. bank, which buys stock in foreign corporations in their domestic currencies and places them with a custodian. The bank then issues dollar ADRs backed by the shares of the foreign stock
- Three main types of ADR issuances
  - *Level 1 ADRs* are the most common and most basic of the ADRs
    - Have the least amount of regulatory requirements
  - *Level 2 ADRs* can be listed on the major stock exchanges, but they have more regulatory requirements than Level 1 ADRs
  - *Level 3 ADRs* represent the most respected ADR level a foreign company can achieve in the U.S. markets

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